

Infrastructure **Evolution**

Promptify Unified Persistence & Isolation Proposal

Branch: EulerOS | Verified: June 2026

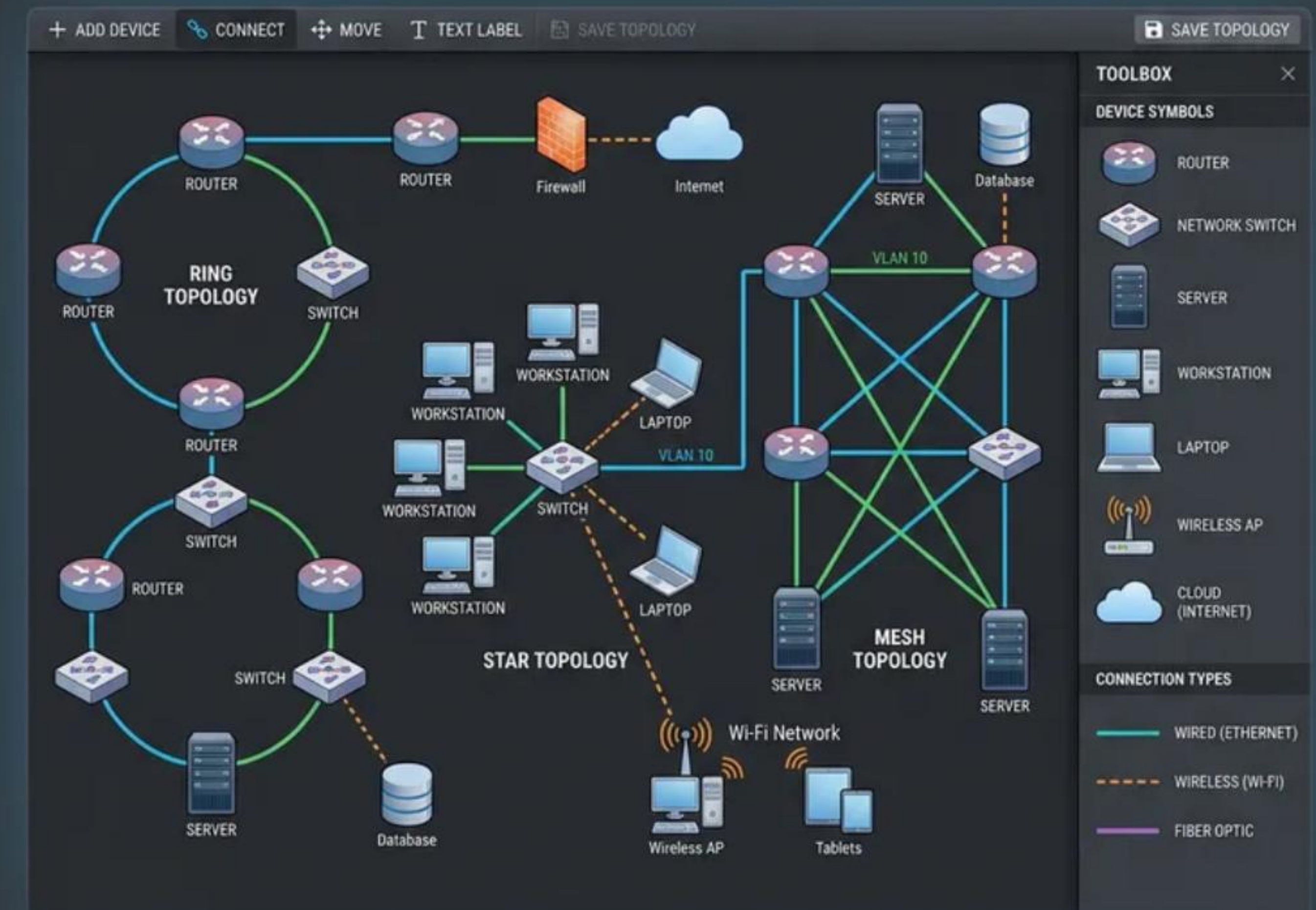
Author: Engineering Lead

INFRASTRUCTURE STATUS AUDIT

The "Dual-Write" Gap

Currently, data lives in `IndexedDB` (primary) and `PostgreSQL` (fallback). These stores diverge silently because of a lack of transactional safety in `useChatHistory.ts:210`.

Technical Debt: Cross-device access is broken. Load-path priority ignores server-side updates if a local cache exists, leading to stale session states.



PROBLEM: RUNTIME DATA VOLATILITY



Current State

Generated apps (CRM, Inventory, HR) store data in **React state only**. Refreshing the browser tab nukes all business records.



The Bottleneck

`SUPABASE_URL` is blank in `.env.prod`. Provisioning logic exists but has no instance to point to. We are shipping "Read-Only" prototypes.

"Scale without persistence is just high-latency experimentation."

PROBLEM: CODEBASE PERSISTENCE

Message Replay

90 Seconds (WASM/NPM Install)

Proposed Snapshot

~4 Seconds
(Direct Mount)

The WASM Execution Tax

Every reload replays `boltAction type="shell"` history. Re-running `npm install` for every user session is an architectural failure that creates a "Broken UI" perception.

Impact: User IDE edits (`files.ts:43`) are ephemeral and lost on tab close.

THE PROPOSED SOLUTION

One Supabase Project Per User

We leverage the Supabase Management API to automatically provision isolated backends at signup.

- **Isolation:** Architectural tenant separation (GDPR ready).
- **Persistence:** Storage buckets for codebase snapshots.
- **Compute:** Dedicated Postgres schemas for generated apps.

Cloud Computing

3D Icon Set

- ☁ 30 3D Icons
- ☁ Layered Objects
- ☁ High-Quality Render
- ☁ Blender, Figma, PSD, PNG



THE ENGINEERING ADDITION

New Component	File Location / Table	Primary Purpose
<code>user_supabase_credentials</code>	<code>database-postgresql.ts</code>	Encrypted Storage of Service Role Keys for provisioning.
<code>codebase_versions</code>	<code>app/lib/database</code>	Metadata for path-based snapshot versioning.
<code>snapshotStorage.server.ts</code>	<code>app/lib/persistence</code>	Core Logic for Save/Load/List operations to Storage.

SYSTEMIC LOGIC CHANGES

Auth Hook

Signups now trigger an async `provisionUserSupabase()` call. User is "live" but infra builds in the background.

Workbench Store

Refactoring `_runAction`. It will now suppress replayed message effects if a snapshot is already mounted.

Visibility Listener

Adding a `visibilitychange` listener to trigger `keepalive: true` fetch calls before tab death.

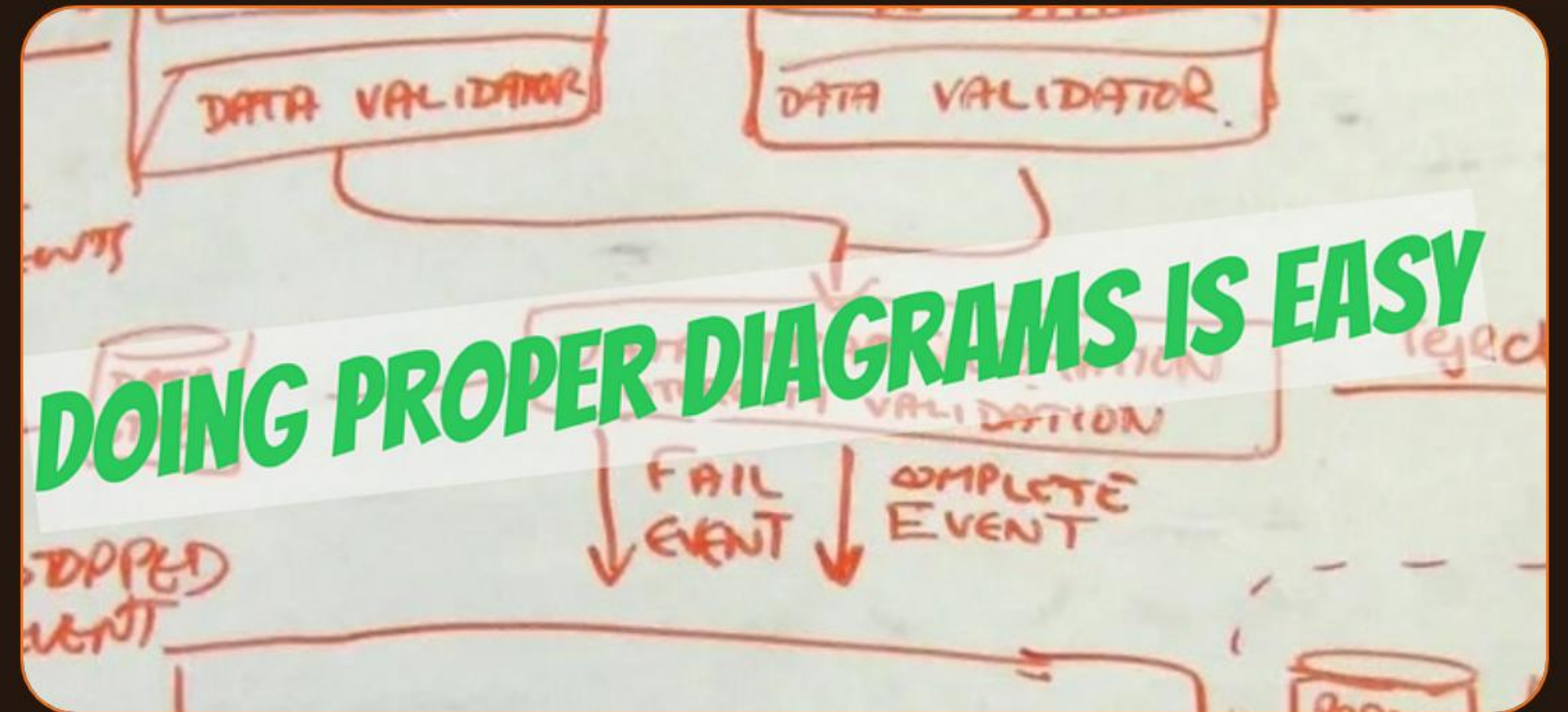
THE SNAPSHOT PROTOCOL

Path-Based Versioning

Supabase Storage does not native-version. We implement:

```
codebase/{chatId}/latest/snapshot.json  
codebase/{chatId}/history/v001_17496.json
```

Build Step: `buildSnapshot()` filters `node_modules` and binary assets to keep payloads <2MB for sub-second REST transfer.

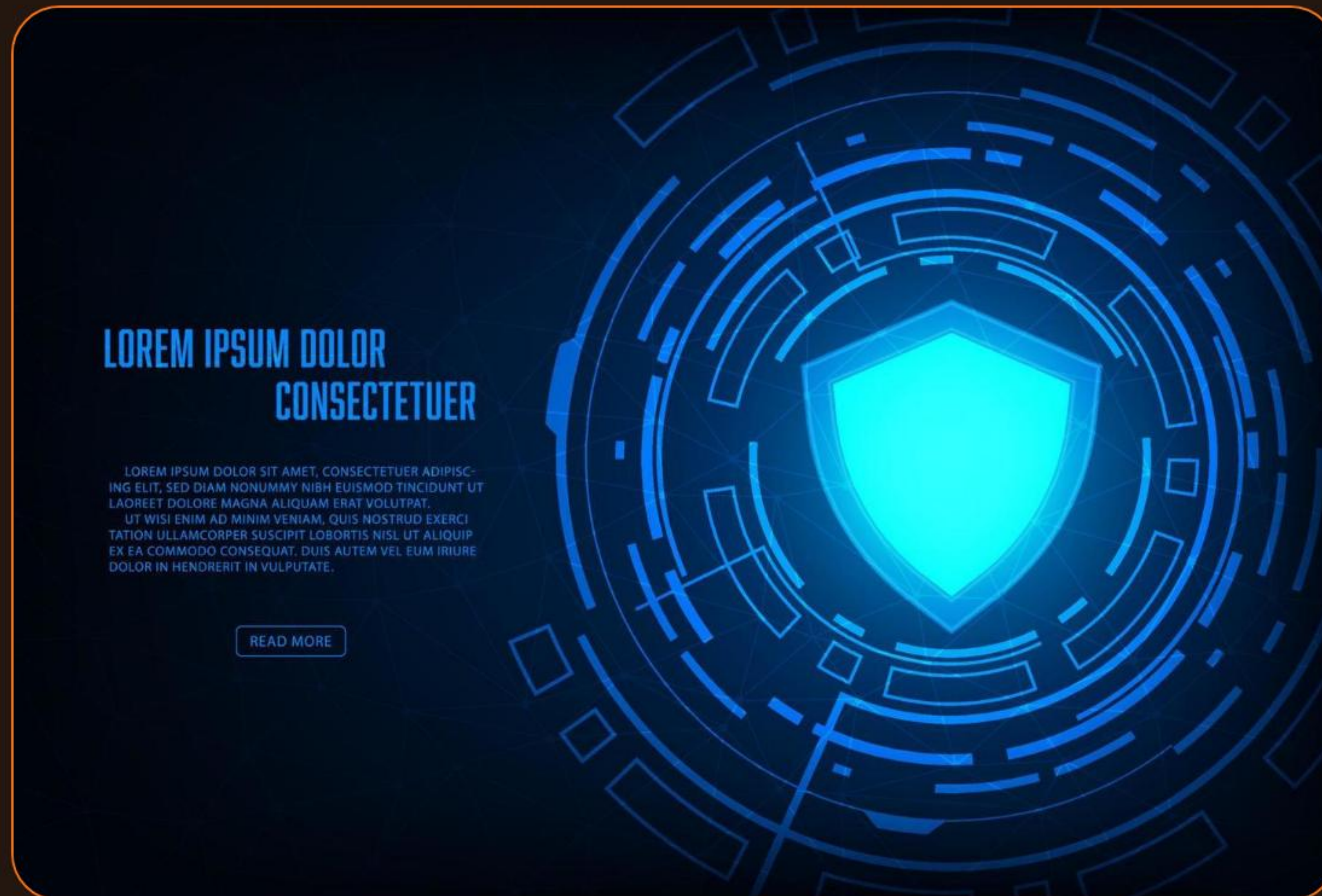


SECURITY & TENANT ISOLATION

Zero Shared Infrastructure

Cross-account isolation is enforced by the **Management API Boundary**.

- Company A and B have separate DB instances.
- Service Role Keys are encrypted at rest with AES-256-GCM.
- Browser never sees the Service Key; only the scoped Anon Key is injected via `env-config.js`.



FINANCIAL JUSTIFICATION

80%

Target Gross Margin

Supabase Pro Base	\$25 / Org
Per-User Project	~\$10 / Mo
Avg. SaaS Revenue	\$49 / User
Gross Margin	~\$34 / User

Note: Pico instances (scale-to-zero) will further reduce idle costs by ~70% once the Supabase for Platforms partnership is approved.

4-WEEK TECHNICAL ROLLOUT

Week 1

Database schema migrations &
Management API connectivity
validation.

Week 2

Snapshot save/load
implementation & Workbench
Store refactor.

Week 3

Per-user provisioning live for new
signups. UI loading overlays
added.

Final Recommendation

Promptify's success as an Enterprise IDE depends on **Persistence** and **Speed**.

I recommend the immediate implementation of the **Isolated Supabase Architecture** to resolve the EulerOS technical debt and enable production-ready app deployments.

Engineering Lead | EulerOS Taskforce

IMAGE SOURCES



Thumbnail
for easy-

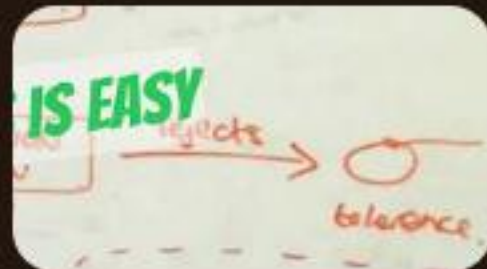
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Source: easy-peasy.ai



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Source: flat-icons.com



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